

Chapter 1- IT Support Introduction



What Does an IT Support Engineer Do?

An **IT Support Engineer** is responsible for **maintaining, troubleshooting, and managing all technical systems** in an organization.



They are called the **“First Line of Defense”** because:

- They solve most problems before escalating to higher teams.
- They ensure business operations do not stop due to IT issues.

Key Responsibilities of IT Support Engineer



Hardware Troubleshooting

What:

Fix physical device issues.

Why:

Hardware failure stops work completely.

Tasks:

- Fix desktops, laptops, printers
- Check cables, ports, power issues

Example:



If a PC is not turning ON → Check power cable, SMPS, motherboard.

2 Operating System Troubleshooting

What:

Fix software/system issues (mainly Windows).

Why:

OS controls entire system.

Tasks:

- Fix slow system
- Resolve login issues
- Handle crashes/errors

Example:

👉 User can't login → Reset password or check domain issue.

3 Network Troubleshooting

What:

Fix internet and connectivity issues.

Why:

Without network → no communication or work.

Tasks:

- LAN/Wi-Fi issues
- Router/Switch problems
- IP conflicts

Example:

👉 No internet → Check cable, IP, router restart.

4 Device Monitoring & Maintenance 🔍

What:

Keep systems healthy.

Tasks:

- Install updates
- Monitor performance
- Run antivirus scans

Example:

👉 System slow → Check CPU/RAM usage.

5 Documentation 📄

What:

Maintain records of issues.

Why:

Helps in tracking & future troubleshooting.

Tasks:

- Ticket logs
 - System info
 - Issue history
-

6 Service Request Handling 🎧

What:

Solve user issues.

Channels:

- Calls 📞
- Emails ✉️

- Ticket tools 🎫
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Devices IT Support Engineer Works With

- Mobile phones 📱
 - Desktops 🖥️
 - Laptops 📖
 - Servers 🖨️
 - Cloud services ☁️
 - Network devices (Router, Switch, Firewall) 🌐
 - Security tools 🔒
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Where IT Support is Needed?

- Airports ✈️
- Malls 🏬
- Hotels 🏨
- Hospitals 🏥
- Manufacturing industries 🏭
- Chemical companies 🧪
- Retail companies 🛒

👉 Example: Billing system down in mall = business loss → IT support fixes immediately.

Important Requirements (Career Start)

1 Learn Office 365

- First 10 lectures minimum
- Email, Teams, OneDrive basics

2 Improve English Communication

 Trick:

- Keep pencil under tongue while speaking
- Improves pronunciation

Common Roles in IT Support

1 Help Desk Support

- First contact person
- Solves basic issues

Example:

 Password reset

2 Desktop Support Engineer

- Works physically on systems

Tasks:

- Install software
 - Setup devices
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3 System Administrator 🖥️

- Manages servers

Tasks:

- Backup
 - Security
 - Active Directory
-

4 Network Administrator 🌐

- Handles networking

Tasks:

- Routers, switches
 - VLAN, firewall
-

5 IT Support Engineer / Technical Support 🛠️

- Handles advanced issues
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◆ Types of Support

Technical Support:

- Handles OS + Network + Server

Product Support:

- Supports company product
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🔒 Cybersecurity Insight

👉 Cybersecurity is a **human psychology game**

- Includes IT Support + Networking
- Many attacks happen due to human mistakes

Hardware Basics – Connectors

Connector Use

24-pin ATX Motherboard power

4/8-pin CPU Processor power

PCIe Graphics card

SATA/Molex Storage devices

SMPS (Switch Mode Power Supply)

What is SMPS?

 Device that supplies power to desktop.

Function:

- Converts **AC (220V)** → **DC (required voltage)**

Why SMPS is Important?

- Computer runs on DC
- House supply is AC
- SMPS converts & stabilizes power

Voltage Distribution

Component Voltage

CPU 12V

RAM 3.3V

USB 5V

Laptop Adapter vs SMPS

Desktop	Laptop
SMPS (internal)	Adapter (external)
Button required	Auto power
Inside CPU cabinet	Charger box

Important Warning

-  Never open SMPS
 -  Capacitor can store **300V** → Dangerous
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Capacitor Role

- Stores current
- Supplies when needed

 Example:
Like water tank storing water

SMPS Wire Color Codes (VERY IMPORTANT)

Color	Voltage	Use
Yellow	+12V	CPU, GPU
Red	+5V	USB
Orange	+3.3V	RAM
Black	Ground	Return
Green	PS_ON	Start signal

Color	Voltage	Use
Purple	5V Standby	Always ON
Gray	Power OK	Status
Blue	-12V	Old ports

How Desktop Power Works

1. Switch ON 
 2. Press Power Button 
 3. Green (PS_ON) connects to Black (GND)
 4. Circuit completes
 5. SMPS sends power
 6. System starts
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Important Concept

-  SMPS does NOT start automatically
 -  Motherboard controls SMPS
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Troubleshooting Scenarios (VERY IMPORTANT)

Scenario 1: Laptop Not Turning ON

Steps:

1. Plug charger
2. Turn switch ON
3. Check voltage using multimeter

 Expected Output: ~18.5V

Result:

- If 10V → Charger issue
 - If OK → Laptop issue
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Scenario 2: Desktop Not Turning ON

Possible Issues:

- SMPS faulty
 - Motherboard faulty
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SMPS Testing (Shorting Method)

Steps:

1. Take metal pin
2. Connect **Green + Black wire**
3. Check fan

 Fan rotates = SMPS working

Interview Question Answer

 Q: How will you troubleshoot no power issue?

Answer:

1. First check power supply
 2. For desktop:
 - Short green & black wire
 3. For laptop:
 - Check adapter voltage (~18.5V)
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⚠ Important Case

👉 If charger LED turns OFF after plugging:

- Indicates **motherboard short circuit**

🧠 Extra Practical Tips (Very Important)

- ✓ Always check **power first**
- ✓ Do not assume → verify
- ✓ Use **multimeter**
- ✓ Follow step-by-step approach
- ✓ Save time → identify root cause fast

🚀 Final Understanding

👉 IT Support = **Knowledge + Troubleshooting Speed**

To become good:

- Learn theory 📖
- Practice real scenarios 🔧
- Improve communication 🗣️

Summary

 **IT Support Engineer kya karta hai?**

 IT Support Engineer ka kaam hota hai:

- System ko **maintain** karna 
- Problems ko **troubleshoot** karna 
- Company ke IT systems ko **smooth chalana**

 Isliye ise kehte hai:

“First Line of Defense” 

Core Responsibilities (Short Summary)

1 **Hardware Troubleshooting**

- Desktop, laptop, printer fix karna
- Power, cable, motherboard check karna

2 **OS Troubleshooting**

- Slow system fix
- Login issue solve
- Windows errors handle

3 **Network Troubleshooting**

- Internet issues
- LAN / Wi-Fi problem
- IP conflict

4 **Monitoring & Maintenance**

- Updates install
- Antivirus run
- Performance check

5 Documentation

- Ticket logs maintain
- Issue history save

6 Service Handling

- Calls, emails, tickets handle
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Devices handled

- Laptop 
 - Desktop 
 - Servers 
 - Network devices 
 - Cloud 
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IT Support kaha use hota hai?

 Har jagah:

- Hospitals 
- Airports 
- Malls 
- Companies 

 Example:

Billing system down = business loss 

IT Support Roles

- Help Desk 
- Desktop Engineer 

- System Admin 📱
 - Network Admin 🌐
 - IT Support Engineer 🔧
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🔒 Cybersecurity Insight

👉 Cybersecurity = **Human + Technology**

- Most attacks = human mistakes
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⚡ Hardware Core Concept (VERY IMPORTANT)

SMPS kya hai?

👉 AC → DC convert karta hai ⚡

Voltage Distribution:

- CPU → 12V
 - RAM → 3.3V
 - USB → 5V
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🎨 SMPS Wire Codes (Important)

- Yellow → 12V
 - Red → 5V
 - Orange → 3.3V
 - Black → Ground
 - Green → Start signal
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🔧 Troubleshooting Basics

👉 Always follow:

- Step-by-step approach

- Check power first
- Use multimeter

Conclusion (Final Understanding)

 IT Support Engineer banne ke liye 3 cheezein sabse important hai:

1. Knowledge

System, OS, Network ka strong base

2. Troubleshooting Skills

Problem ko fast identify karna

3. Communication

User se clear baat karna

 Final line:

IT Support = Knowledge + Speed + Practical Skills 

Q & A

BASIC QUESTIONS

 **Q1: What is IT Support?**

 **Answer:**

IT Support is responsible for maintaining, troubleshooting, and managing IT systems in an organization.

They ensure that systems run smoothly and users face minimal downtime.

 **Q2: Why IT Support is called First Line of Defense?**

 **Answer:**

Because IT Support solves most issues at the initial level before escalating to higher teams.

This helps in saving time and maintaining business continuity.

? Q3: What are your key responsibilities?

👉 Answer:

- Hardware troubleshooting
- OS troubleshooting
- Network troubleshooting
- User support
- Documentation

◆ **HARDWARE QUESTIONS**

? Q4: What is SMPS?

👉 Answer:

SMPS (Switch Mode Power Supply) converts AC power into DC power required by computer components.

? Q5: Why SMPS is important?

👉 Answer:

Because computer components work on DC power, while our home supply is AC.

SMPS converts and stabilizes voltage.

? Q6: What is capacitor?

👉 Answer:

A capacitor stores electrical energy and supplies it when required.

? Q7: What are SMPS wire colors?

👉 Answer:

- Yellow → 12V
 - Red → 5V
 - Orange → 3.3V
 - Black → Ground
-

◆ OS QUESTIONS

? Q8: How to fix slow system?

👉 Answer:

- Check CPU/RAM usage
 - Remove startup programs
 - Run antivirus
 - Disk cleanup
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? Q9: User cannot login, what will you do?

👉 Answer:

- Check password
 - Reset password
 - Check domain connection
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◆ NETWORK QUESTIONS

? Q10: No internet issue – how will you troubleshoot?

👉 Answer:

Step-by-step:

1. Check cable
2. Check IP address
3. Restart router
4. Ping test

? Q11: What is IP conflict?

👉 Answer:

When two devices have same IP address → network issue occurs.

◆ PRACTICAL SCENARIOS (VERY IMPORTANT)

💻 Scenario 1: Laptop not turning ON

👉 Answer:

1. Check charger
 2. Check power switch
 3. Measure voltage (~18.5V)
 4. If low → charger issue
 5. If OK → motherboard issue
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Scenario 2: Desktop not turning ON

Answer:

- Check SMPS
 - Check motherboard
 - Use shorting method
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Scenario 3: How to test SMPS?

Answer:

- Connect Green + Black wire
 - Check fan rotation
 -  If fan runs = SMPS working
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Scenario 4: Charger LED turns OFF

Answer:

 Indicates motherboard short circuit

Q12: How do you handle angry user?

Answer:

- Stay calm
 - Listen carefully
 - Assure quick resolution
 - Give proper update
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Q13: What is ticketing system?

Answer:

A system used to track user issues and manage support requests.

? Q14: Difference between Helpdesk and Desktop Support?

👉 Answer:

Helpdesk	Desktop Support
Remote support	Physical support
Basic issues	Advanced issues

? Q15: How do you troubleshoot quickly?

👉 Answer:

- Follow structured approach
- Identify root cause
- Eliminate possibilities
- Use tools

? Q16: What is most important skill?

👉 Answer:

👉 Troubleshooting + Communication
